

Recall from last class:

- An **orthogonal matrix** U is a square matrix so that $U^{-1} = U^T$.
- **Orthogonal Decomposition Theorem:** Let W be a subspace of $\mathbb{R}^n$. Then each $y \in \mathbb{R}^n$ can be uniquely written as

$$y = \hat{y} + z$$

where $\hat{y} \in W$ and $z \in W^\perp$.

Moreover, if $\{u_1, \dots, u_p\}$ is any orthogonal basis for W , then

$$\hat{y} = \left(\frac{y \cdot u_1}{u_1 \cdot u_1} \right) u_1 + \dots + \left(\frac{y \cdot u_p}{u_p \cdot u_p} \right) u_p,$$

and $z = y - \hat{y}$.

- The vector $\hat{y}$ is the **orthogonal projection** of y onto the subspace W . Written as

$$\hat{y} = \text{proj}_W y.$$

- **Best Approximation Theorem:** Let W be a subspace of $\mathbb{R}^n$. Let $\hat{y}$ be the orthogonal projection of y onto W , then

$$\|y - \hat{y}\| < \|y - v\|$$

for all $v \in W \setminus \{\hat{y}\}$.

- **Theorem 6.10:** If $\{u_1, \dots, u_p\}$ is an orthonormal basis for W , then

$$\text{proj}_W y = (y \cdot u_1)u_1 + \dots + (y \cdot u_p)u_p.$$

If $U = [u_1 \dots u_p]$, then

$$\text{proj}_W y = UU^T y.$$

- **Gram-Schmidt Orthogonalization.** Given a basis $\{x_1, \dots, x_p\}$ for a non-zero subspace W of $\mathbb{R}^n$, define

$$\begin{aligned} v_1 &= x_1 \\ v_2 &= x_2 - \left(\frac{x_2 \cdot v_1}{v_1 \cdot v_1} \right) v_1 \\ v_3 &= x_3 - \left(\frac{x_3 \cdot v_1}{v_1 \cdot v_1} \right) v_1 - \left(\frac{x_3 \cdot v_2}{v_2 \cdot v_2} \right) v_2 \\ &\vdots \\ v_p &= x_p - \left(\frac{x_p \cdot v_1}{v_1 \cdot v_1} \right) v_1 - \dots - \left(\frac{x_p \cdot v_{p-1}}{v_{p-1} \cdot v_{p-1}} \right) v_{p-1}. \end{aligned}$$

Then $\{v_1, \dots, v_p\}$ is an orthogonal basis for W . In addition $\text{Span}\{v_1, \dots, v_k\} = \text{Span}\{x_1, \dots, x_k\}$ for all $1 \leq k \leq p$.

- **QR Factorization.** If A is an $m \times n$ matrix with linearly independent columns, then $A = QR$ where Q is an $m \times n$ matrix whose columns form an orthonormal basis for the column space of A and R is an upper triangular $n \times n$ matrix with positive entries along the diagonal.

Goal: Solve the matrix $Ax = b$.

What if the matrix equation is inconsistent?

Definition 1. A **least squares solution** to $Ax = b$ is a vector $\hat{x} \in \mathbb{R}^n$ such that

$$\|b - A\hat{x}\| \leq \|b - Ax\|$$

for all $x \in \mathbb{R}^n$.

How can we find least squares solutions?

Theorem 6.13: The set of least squares solutions to $Ax = b$ is the same as the set of solutions to the normal squares $A^T Ax = A^T b$.

1. Prove the theorem.

Definition 2. *The least squares error is given by the distance from b to $A\hat{x}$.*

2. Let $A = \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}$ and $b = \begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix}$. Find a least squares solutions to $Ax = b$.

3. Let $A = \begin{bmatrix} 1 & -6 \\ 1 & -2 \\ 1 & 1 \\ 1 & 7 \end{bmatrix}$ and $b = \begin{bmatrix} -1 \\ 2 \\ 1 \\ 6 \end{bmatrix}$. Find a least squares solutions to $Ax = b$.

4. Find the best fit line through the points $(-6, -1)$, $(-2, 2)$, $(1, 1)$, $(7, 6)$.

5. Here are the number of daylight hours in Cambridge on a few days of 2019, from the Astronomical Applications Department of the U.S. Naval Observatory.

Day	Which day of the year	number of daylight hours
May 17	137th	14 hours and 40 minutes
June 1	152nd	15 hours and 5 minutes
July 1	182nd	15 hours and 14 minutes
August 8	220th	14 hours and 12 minutes

Suppose you would like to fit a function f that gives the number of hours of daylight on a particular day, so that you can predict the number of daylight hours on the shortest day of the year. Describe how you would go about this; what form would your function f have?

Least Squares – Answers / Solutions

1. *Proof.* We prove the two sets are equal by showing that each is a subset of the other.

First, let $\hat{x}$ be a least squares solution to $Ax = b$. Let $\hat{b} = \text{proj}_{\text{Col}(A)}b$. Thus $A\hat{x} = \hat{b}$. By the Orthogonal Decomposition Theorem, $b - \hat{b}$ is orthogonal to $\text{Col}(A)$, and hence $b - \hat{b}$ is orthogonal to the columns a_j of A . Thus we have $a_j^T(b - A\hat{x}) = 0$ for $1 \leq j \leq n$. Hence $A^T(b - A\hat{x}) = 0$, and rearranging we get $A^Tb = A^TA\hat{x}$. Thus $\hat{x}$ is a solution to the normal equations.

Second, suppose $\hat{x}$ is a solution to the normal equations. Rearranging the normal equations, we get

$$A^T(b - A\hat{x}) = 0.$$

Thus $b - A\hat{x}$ is orthogonal to the column space of A . Since $b = A\hat{x} + (b - A\hat{x})$ is an orthogonal decomposition of b and the orthogonal decomposition is unique, we know $A\hat{x} = \text{proj}_{\text{Col}(A)}b$. By the Best Approximation Theorem, $\hat{x}$ is a least squares solution. $\square$

2. We use the normal equations to find the least squares solutions. Note

$$A^TA = \begin{bmatrix} 17 & 1 \\ 1 & 5 \end{bmatrix} \quad \text{and} \quad A^Tb = \begin{bmatrix} 19 \\ 1 \end{bmatrix}.$$

Thus

$$\hat{x} = \begin{bmatrix} 17 & 1 \\ 1 & 5 \end{bmatrix}^{-1} \begin{bmatrix} 19 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

3. Note the columns of A are orthogonal. Now

$$\hat{b} = \text{proj}_{\text{Col}(A)}b = \frac{b \cdot a_1}{a_1 \cdot a_1}a_1 + \frac{b \cdot a_2}{a_2 \cdot a_2}a_2 = \frac{8}{4}a_1 + \frac{45}{90}a_2 = \begin{bmatrix} -1 \\ 1 \\ 5/2 \\ 11/2 \end{bmatrix}.$$

Now we need to solve $Ax = \hat{b}$, and hence we get $\hat{x} = \begin{bmatrix} 2 \\ 1/2 \end{bmatrix}$.

4. We fit the data to the line $y = \beta_0 + \beta_1x$, and hence we get the following four equations.

$$\begin{aligned} -1 &= \beta_0 + \beta_1(-6) \\ 2 &= \beta_0 + \beta_1(-2) \\ 1 &= \beta_0 + \beta_1(1) \\ 6 &= \beta_0 + \beta_1(7) \end{aligned}$$

Thus we can use the matrix A and the vector b from the last problem. The least squares solution is then $\begin{bmatrix} 2 \\ 1/2 \end{bmatrix}$. Thus the best fit line is $y = 2 + \frac{x}{2}$.

5. We can try to fit the data to the function

$$f(t) = c_0 + c_1 \cos\left(\frac{2\pi}{365}t\right) + c_2 \sin\left(\frac{2\pi}{365}t\right).$$